ACHELION CAPITAL MANAGEMENT, LLC

Understanding ARMS

The Achelion Risk Management System

A Plain-Language Briefing for General Partners

March 2026 · Confidential — For General Partners Only

> *Most people love cars without knowing how combustion works. This document explains the car. The engineering manuals exist — but this is not one of them.*

A Note to the General Partners

ARMS — the Achelion Risk Management System — is the decision-making engine that runs Achelion Capital. It determines how much risk we take, when we take it, when we pull back, and how we protect capital when markets break. It does not replace judgment. It is the architecture that ensures judgment is applied consistently, without the distortions of fear, greed, fatigue, or overconfidence.

This document is written for you — the General Partners who represent Achelion to the world, who sit across the table from LPs, and who need to understand what we have built well enough to describe it with conviction. You do not need to know the code. You do need to know the logic.

Think of this as learning to fly a commercial aircraft as a passenger rather than a pilot. You will never touch the controls. But you should know what the autopilot does, why it exists, and why you can trust it more than a tired human at the controls at 35,000 feet.

> *ARMS is not a trading bot. It is a discipline engine. It ensures that the decisions Achelion makes are always the decisions Achelion intended to make — regardless of market conditions, time of day, or emotional climate.*

Glossary — Terms You Will Encounter

Every term below appears in this document. Read this section once. Return to it as needed.

ARMS	Achelion Risk Management System. The complete decision and execution architecture that governs how Achelion manages capital. Not a single tool — a system of seven interconnected modules.
Architecture AB	The canonical portfolio structure: 58% equities, 20% digital assets, 14% defensive sleeve, 8% cash. 'AB' stands for Achelion Base.
Regime	The market environment we are currently operating in. ARMS identifies five regimes — from most aggressive to most defensive. The regime determines how much risk we are allowed to carry.
Regime ceiling	The maximum equity exposure allowed in each regime. In the most defensive regime, we can only hold 15% in equities. In the most aggressive, we can hold 100%. The ceiling is automatic and cannot be overridden without documented GP approval.
Conviction score	A number from 1 to 10 that represents our confidence in a position. Higher conviction = larger position size. The formula squares the score, so a C10 position carries 4 times the weight of a C5. This is mathematical concentration discipline — bet big on what you know, small on what you are learning.
MICS	Model-Implied Conviction Score. The formula ARMS uses to derive the conviction score from objective data — rather than relying on the PM's subjective opinion. The PM can adjust it by one level with a written reason. That is the full extent of PM involvement in sizing.
SENTINEL	The six-gate protocol that every new investment must pass before entering the portfolio. Think of it as a six-question interview that every position must answer correctly before we invest a dollar.

Defensive sleeve	14% of the portfolio permanently allocated to instruments that hold value or gain value when markets fall: short-term Treasury bills (SGOV), gold (SGOL), managed futures (DBMF), and structured credit (STRC). The sleeve is not a hedge — it is structural protection that is always present.
PTRH	Permanent Tail Risk Hedge. We always hold put options on the Nasdaq-100 ETF (QQQ). These options pay off if technology stocks fall sharply. Always on. Never removed. Think of it as permanent insurance.
SLOF	Synthetic Leverage Overlay Facility. A tool that allows us to amplify returns on our highest-conviction positions without borrowing money. Only used in the two most aggressive market regimes. Automatically deactivated when markets deteriorate.
PERM	Profit-locking mechanism. When a position has risen more than 30% from our entry price, ARMS evaluates selling covered call options against it. This collects cash premium from other investors and provides a buffer if the position reverses. Protects gains without selling the position.
CDF	Conviction Decay Function. If a position underperforms the Nasdaq-100 by more than 10 percentage points for 45 consecutive days, its weight in the portfolio automatically begins to shrink — without requiring the PM to make a sell decision. At day 135 without improvement, a mandatory review is triggered.
ARAS	Achelion Risk Assessment System. The module that continuously scores the current market environment and outputs the current regime and ceiling. The brain of ARMS.
PDS	Portfolio Drawdown Sentinel. An independent safety mechanism that watches the portfolio's total value against its all-time high. If we fall 12% from our peak, exposure automatically reduces to 60%. If we fall 18%, it reduces to 30%. This fires regardless of what any other module says.
FEM	Factor Exposure Monitor. Detects hidden concentration risk. Our portfolio may look diversified across 12 companies — but if 80% of those companies share the same dependence on AI chip manufacturing in Taiwan, we are effectively running a single concentrated bet. FEM surfaces this.

Session log	The immutable record of every decision ARMS makes. Every trade, every regime change, every flag, every override — logged with a timestamp, the reason, and the module that triggered it. Cannot be edited. LP-auditable.

Tier 0 / Tier 1 / Tier 2	The three levels of decision authority. Tier 0: the system acts immediately with no human input. Tier 1: the system acts unless the PM vetoes within a defined window. Tier 2: the system recommends, a human must confirm. No decision can be escalated from Tier 0 to Tier 1 without GP documentation.

What ARMS Does — In One Paragraph

> *ARMS watches markets continuously, identifies the current environment, sets the maximum risk we are allowed to take, sizes every position according to a formula rather than opinion, protects capital automatically when things deteriorate, harvests gains from winning positions, and generates a complete daily report — all without human involvement. The PM's job is to identify which companies belong in the portfolio. Everything after that decision is ARMS.*

The analogy we use internally: ARMS is the flight control computer on an F-22 fighter jet. The pilot — our PM — declares the mission and identifies the target. The computer translates that intent into precise control surface movements the human could never execute fast enough or accurately enough. The pilot does not manage individual aileron deflections. The pilot flies the mission. The computer flies the jet.

What this means for our LPs: they are not betting on our PM's ability to make good decisions every day under pressure. They are betting on an architecture that was specifically designed to perform when markets are most difficult — because that is precisely when human judgment is most compromised.

The Five Regimes — Reading the Room

ARMS identifies the current market environment every five minutes and classifies it into one of five regimes. The regime is not a forecast — it is a present-tense assessment of current conditions. Think of it as ARMS continuously asking: 'Given what the market is doing right now, how much should we have at risk?' The answer changes the portfolio ceiling automatically.

The regime is determined by a composite score derived from multiple signals: growth momentum, inflation trends, oil prices, interest rate trajectory, credit conditions, volatility levels, and geopolitical event risk. No single signal controls the outcome — it is a weighted combination, like a pilot reading multiple instruments simultaneously.

Regime	Score range	Equity ceiling	SLOF status	What it means
RISK_ON	0.00 – 0.30	**100%**	Active — full deployment	Everything is working. Growth is strong, markets are calm, risk is rewarded. We are fully invested and using our conviction amplifier.
WATCH	0.31 – 0.50	**100%**	Active — monitoring	Conditions are mostly positive but something is worth watching. We remain fully invested but the system is on alert. Defense is beginning to build.
NEUTRAL	0.51 – 0.65	**75%**	Reducing	The environment is mixed. We pull back to 75% equity exposure. The conviction amplifier begins to reduce. We are cautious but not defensive.
DEFENSIVE	0.66 – 0.80	**40%**	Removed	Conditions are deteriorating. We hold a maximum of 40% in equities. The conviction amplifier is off. The defensive sleeve and tail hedge are working. We are protecting capital.
CRASH	0.81 – 1.00	**15%**	Removed	Markets are in freefall. We hold a maximum of 15% in equities. Everything else is in capital preservation mode. No new positions. No amplification. Survival first.

One important nuance: the regime is not a forecast. ARMS does not predict where the market is going. It identifies where it is right now and manages risk accordingly. This is a crucial distinction. Many funds try to predict the future. ARMS responds to the present — faster and more accurately than any human can.

The Seven Pillars — How ARMS Is Built

ARMS is not a single system. It is seven interconnected modules — we call them pillars — each responsible for a specific function. Together they form a complete architecture. Remove any one and the system has a gap. All seven must operate simultaneously for ARMS to function as designed.

Think of the seven pillars as the seven systems on a commercial aircraft: engines, navigation, hydraulics, pressurization, electrical, communications, and landing gear. Each does something distinct. Each depends on the others. The plane does not fly on six.

1

ARAS · Achelion Risk Assessment System · *The brain.*

ARAS continuously reads the market — growth data, inflation signals, oil prices, interest rates, credit conditions, volatility — and produces the current regime label and ceiling. Every other module takes its instructions from ARAS. It runs every five minutes, twenty-four hours a day. If markets move at 3am, ARAS knows.

> **Analogy:** *The air traffic controller. Constantly scanning the environment, telling the pilot what altitude to fly at given current conditions.*

2

Macro Compass · Regime Detection Engine · *The navigator.*

The Macro Compass is the sub-system inside ARAS that converts raw market data into the regime score. It weighs multiple signals and produces a single number between 0 and 1. Below 0.50 is the optimistic half. Above 0.65 is the defensive zone. The score tells us not just which regime we are in — but where we sit within that regime, how close we are to a boundary, and which direction we are moving.

> **Analogy:** *The instrument panel. Not just showing speed — showing rate of acceleration, fuel consumption, and distance to the next waypoint simultaneously.*

3

Master Engine · Portfolio Construction · *The builder.*

The Master Engine takes the regime ceiling from ARAS and the conviction scores from MICS, and calculates the precise target weight for every position. It enforces the ceiling automatically. It applies the conviction-squared weighting formula. It incorporates any conviction decay factors from underperforming positions. The PM does not set position sizes — the Master Engine calculates them.

> **Analogy:** *The autopilot's flight management computer. Given the destination, altitude constraints, and fuel load — it calculates the precise control inputs required. The pilot does not manually calculate thrust settings.*

4

KEVLAR · Position Limits · *The guardrails.*

KEVLAR enforces hard maximum position sizes regardless of what any other module says. No single position can grow beyond its KEVLAR limit — not because of conviction, not because of performance, not because the PM believes in it. KEVLAR exists to prevent the catastrophic scenario where one position becomes so large that it threatens the entire portfolio. It is the structural steel in the building.

> **Analogy:** *The speed limiter on a car. No matter how hard you press the accelerator, the car will not exceed 155mph. The driver's intention is irrelevant. The limit is structural.*

5

PERM · Profit Protection · *The harvester.*

When a position has risen more than 30% from our entry price, PERM evaluates whether to sell covered call options against it. If it does, we collect cash from other investors who want to bet the position will keep rising sharply. That cash is ours regardless of what happens next. If the position keeps rising, we participate — but less. If it falls, the cash we collected cushions the impact. PERM also manages the Conviction Decay Function, which automatically reduces the size of positions that are underperforming.

> **Analogy:** *The farmer who sells a portion of next season's crop at today's prices once they have a good harvest. They lock in some of the gain. They still benefit from further growth. They protect against a bad season.*

6

SLOF · Conviction Amplifier · *The accelerator.*

SLOF is the only module that adds risk rather than removing it. When we are in RISK_ON or WATCH regime and we have a position at our highest conviction level (C10), SLOF allows us to amplify the return on that position using a synthetic structure that mimics owning double the shares. It is only deployed on our absolute highest-conviction positions, only in the most favorable market conditions, and is automatically removed the moment conditions deteriorate. It is a precision tool, not a lever we pull freely.

> **Analogy:** *The afterburner on a fighter jet. Extraordinary additional power available only in specific conditions. The pilot does not fly with it constantly — it burns fuel too fast. It is reserved for the moments when maximum performance is required and the conditions support it.*

7

ARES · Re-entry System · *The re-builder.*

When ARMS has deleveraged — reduced exposure because conditions deteriorated — ARES manages the return to full deployment. It checks four conditions before allowing any increase in exposure: the regime must have improved, volatility must be declining, the specific trigger that caused the deleverage must have resolved, and the composite risk score must have fallen sufficiently. Re-entry happens in three tranches over several days. We never go back to full exposure in one movement. We test the ground before putting full weight on it.

> **Analogy:** *The aircraft landing system that requires multiple conditions to be met before touchdown: correct speed, correct altitude, correct approach angle, landing gear confirmed down. Not one condition — all four. In sequence.*

The SENTINEL Protocol — How We Choose What We Own

Every position in the Achelion portfolio must pass six questions before we invest a dollar. We call this the SENTINEL protocol. It is the most important discipline in the system — because the best risk management begins with owning the right things in the first place.

Most funds ask one question: will this go up? SENTINEL asks six. Each question must be answered correctly. A single failure means the position does not enter the portfolio — regardless of how compelling the story sounds, how many analysts recommend it, or how much conviction the PM has.

> *A compelling story is exactly what SENTINEL is designed to resist. History is full of investments that had a compelling story and destroyed capital. The gates exist to separate conviction from narrative.*

Gate 1 · Is this a civilizational shift — not a product cycle?

Does this company sit inside a change that will restructure the global economy regardless of whether any individual product succeeds? We only invest in companies that are non-optional components of the way the future will work. We do not invest in trends. We invest in infrastructure.

> **Example:** *We would not have invested in any individual social media app. We would have invested in the companies that built the cloud infrastructure every app ran on.*

Gate 2 · Can the world route around this company?

If this company ceased to exist tomorrow, could the market replace what it does within 18 months? If yes — the company is replaceable and its pricing power is limited. We need companies that cannot be routed around. Structural necessity, not temporary advantage.

> **Example:** *NVIDIA cannot currently be routed around for certain AI computing tasks. A company that makes generic memory chips can be replaced by any of ten competitors. One passes Gate 2. The other does not.*

Gate 3 · Is the market pricing this incorrectly?

This gate uses a quantitative formula — not an opinion. It scores the company's valuation relative to similar companies, the gap between where large institutional investors are positioned and where the thesis says they should be, and whether the investment narrative the market is using to value this company is the wrong one. The score must exceed a defined threshold. A compelling argument does not substitute for the number.

> **Example:** *Alphabet (Google) was valued as an advertising company while becoming one of three sovereign-scale AI computing platforms. The market was using the wrong identity to price it. That gap is what Gate 3 detects.*

Gate 4 · Does this position make the portfolio safer or more dangerous?

The portfolio already holds 12 positions. Adding a 13th changes the concentration of risk across the whole book. Gate 4 runs a factor analysis — does this new position increase our dependence on a single theme (like AI chip manufacturing in Taiwan) beyond safe limits? If it does, we cannot add the position without simultaneously reducing another.

> **Example:** *You would not add a 13th passenger to a 12-passenger boat without first checking whether the hull can handle the additional weight — and whether adding that passenger changes the balance in a way that makes the boat less stable.*

Gate 5 · Is this the right time?

This gate is fully automated. The current market regime determines whether we are allowed to initiate new positions. In DEFENSIVE regime, new positions are blocked and queued until conditions improve. In CRASH regime, nothing new enters the portfolio. The system will not override this gate regardless of how attractive the opportunity looks. Timing discipline is not optional.

> **Example:** *You do not launch a rocket into a thunderstorm, regardless of how perfect the payload is or how long you have been preparing. You wait for the right launch window.*

Gate 6 · What do you know and how do you know it?

This is the only human gate in the protocol. If the PM is bringing a new idea to the table, they must declare the source of their insight from a defined list: a direct conversation with someone inside the company or supply chain (highest value), recognition of a pattern from prior market cycles (high value), or synthesis of publicly available information (lower value — requires a higher score on Gate 3 to compensate). Gut feelings and strong opinions are not a valid source category. They do not exist in the system.

> **Example:** *A surgeon who says 'I have a feeling about this diagnosis' is held to a different standard than one who says 'I spoke with the patient's referring physician and reviewed the prior imaging.' Both may be right. One is defensible. One is not.*

The Daily Flow — What Happens Every Day

ARMS does not sleep. Markets move twenty-four hours a day. Geopolitical events happen at 3am. Oil prices shift before New York opens. ARMS processes all of it continuously. Below is the flow of a typical day.

All night	**Data ingestion**	ARMS continuously monitors market signals, macro data, oil prices, volatility indicators, and geopolitical feeds. The regime score updates every five minutes.
5:15 AM CT	**Portfolio snapshot**	ARMS takes a precise reading of every position — current price, weight, conviction score, decay status, unrealized gain/loss. This is the baseline for the day.
5:20 AM CT	**Module sweep**	All seven pillars run simultaneously: factor exposure check, drawdown sentinel check, tail hedge status, conviction decay update, stress scenario computation.
5:30 AM CT	**Daily monitor compiled**	ARMS assembles the complete daily report — regime assessment, position status, all flags, all alerts, decision queue — and generates it as a digital document.
6:00 AM CT	**Monitor delivered**	The PM and all General Partners receive the daily monitor. No human assembled it. No human edited it. It reflects the state of the portfolio exactly as ARMS sees it.
Market open (9:30)	**Live execution mode**	Any Tier 0 actions — regime ceiling enforcement, position size adjustments, tail hedge rolls — execute automatically as market conditions warrant. No human required.
During market hours	**Continuous monitoring**	ARMS watches every position, every macro signal, and every module trigger in real time. If something requires action, it acts (Tier 0) or notifies the PM (Tier 1) immediately.
Weekly (Monday)	**Scan Engine runs**	ARMS automatically screens the AI infrastructure investment universe — 50 to 75 companies — against the first two SENTINEL gates. Candidates that pass are surfaced to the PM as an intelligence briefing.
Monthly (1st)	**Learning loop runs**	ARMS reviews its own decisions from the prior month. Were the positions it flagged for concern actually problems? Did its conviction scores predict outcomes? The system scores itself and adjusts accordingly.

What Humans Do — And What They Do Not

One of the questions GPs are most likely to hear from LPs is: 'If the system does all of this, what does the PM actually do?' This is the right question. The answer is important.

> *The PM does the one thing no algorithm can replicate: he sees the future before it is in the data. Everything else is the system.*

Achelion's PM brings 28 years of market pattern recognition — the ability to identify a civilizational shift before it reaches mainstream consensus and before the data confirms it. The Tesla thesis. The AI infrastructure thesis. These were not data conclusions. They were pattern recognitions. A human who has seen multiple market cycles can identify when a company is being valued for what it used to be rather than what it is becoming. That is not a skill algorithms currently possess.

Once the PM identifies a candidate and brings it through the SENTINEL protocol, the system takes over. Here is the precise boundary:

The PM does this	The PM does not do this
Identifies companies that represent civilizational shifts before the market prices them correctly	Set position sizes — the MICS formula does this
Brings pattern recognition from lived experience across multiple market cycles	Execute trades — the execution layer does this automatically
Seeds new SENTINEL analyses with a thesis — why this company, why now	Decide when to deleverage — ARAS and PDS do this automatically
Declares the source of the insight (direct knowledge, pattern recognition, or public synthesis)	Override the regime ceiling — this requires documented GP co-sign
Reviews and confirms or vetoes Tier 1 system recommendations within defined time windows	Decide when to re-enter — ARES manages re-entry in tranches
Builds and maintains relationships with LPs and the investment community	Manage the tail hedge — PTRH rolls and resizes this automatically
Holds the strategic vision for Achelion Capital	Make gut-feel decisions under pressure — the system exists precisely to prevent this

What Makes Achelion Different — The Honest Answer

Every hedge fund tells LPs they have a disciplined process. What they mean is that their PM has good intentions and tries to be disciplined. ARMS makes the discipline structural. The difference is the difference between a driver who tries not to speed and a car with a speed limiter. One depends on willpower under pressure. The other does not.

Here is what Achelion can say that almost no other fund of any size can say truthfully:

> **Every sizing decision is derived from a formula, not an opinion.**
> The MICS formula converts SENTINEL gate data into a model-implied conviction score. The PM can adjust it by one level with a written reason. That is the full extent of human involvement in position sizing. No other emerging fund operates this way.

> **The regime ceiling cannot be overridden by the PM alone.**
> If ARMS says we are in DEFENSIVE regime and our equity ceiling is 40%, the PM cannot change that unilaterally. A GP co-sign and documented justification are required. This is not a policy. It is structural — the system enforces it.

> **Every decision is logged with the reason it was made.**
> The session log records every trade, every regime change, every flag, every override. It cannot be edited. An LP, auditor, or regulator can verify that the system ran exactly as described in any document we have provided — in any session, on any date.

> **The defensive architecture was validated through the worst tech year since 2000.**
> The 2022 backtest shows ARMS at –16.1% in a year when the Nasdaq-100 fell 32.6%. The defensive sleeve, the tail hedge, and the regime-driven deleverage all performed as designed when it mattered most. We did not project this. We demonstrated it.

> **The PM's job is to see the future — not to manage the process.**
> Most fund managers spend the majority of their time on portfolio mechanics — what to buy, when to buy, how much, when to sell. ARMS handles all of that. Our PM spends time on the one thing that produces alpha: identifying what the world will look like before the market prices it in.

Questions GPs Will Hear — Suggested Responses

The following questions are the ones LPs ask most often about systematic and AI-assisted portfolio management. These are not scripted answers — they are frameworks for honest responses that build trust.

"What happens when the system is wrong?"

ARMS is not designed to be always right. It is designed to be wrong in ways that are survivable. When the regime model misreads the environment — and it will, occasionally — the structural safeguards still apply. The tail hedge is always on. The drawdown sentinel fires at –12%. The defensive sleeve is always present. The system fails gracefully. Most funds fail catastrophically, because there is no architecture — just a PM who was wrong.

"What if the AI makes a bad decision?"

ARMS does not make decisions. It enforces rules. There is no AI deciding which stock to buy. There is a formula that calculates a conviction score. There is a protocol that gates every new position through six questions. There is a scheduler that enforces ceilings automatically. The intelligence of the system is in its design — not in real-time discretion. When we say AI, we mean the analytical layer that helps assess investment candidates and generate reports — not an autonomous agent trading our capital.

"What does the PM actually do?"

The PM does the one thing no algorithm can currently replicate: he sees the future before it is in the data. His job is to identify companies that the market is pricing incorrectly because it is using the wrong frame of reference. Once he identifies a candidate, the system evaluates it through six gates. If it passes, the system manages it from entry through exit. The PM's job is vision. The system's job is execution.

"How is this different from a quant fund?"

Quant funds find patterns in historical data and bet that those patterns will repeat. We do not trade patterns. We invest in the companies that are building the next era of global infrastructure — and we use the system to ensure we stay invested in them correctly, protect capital when markets break, and do not make emotional decisions under pressure. The thesis is human. The discipline is systemic.

"What is your track record?"

Achelion has not yet begun live fund operations. What we have is a full-cycle backtest — a three-year simulation (2020-2022) of ARMS applied to actual historical market conditions, covering the fastest bear market in history, the most speculative bull run of our era, and the

worst year for technology stocks since 2000. The result: +47.4% over three years versus the S&P 500's +24.8%. The 2022 preservation result was −16.1% versus −18.1% for the S&P 500 — in a year when we carried a 20% allocation to digital assets that fell 65%. The system worked when it was hardest. We show you the full cycle — not our best year.

"Can I see the system running?"

Yes. Every morning, the system generates a daily monitor that shows the current regime, every position with its status and flags, all active alerts, and the decision queue. It is assembled automatically, without human involvement, and reflects the exact state of the portfolio. We are happy to share a sample monitor during due diligence. You will see the architecture working — not a pitch deck claiming it works.

In Summary

ARMS is not a feature. It is the fund. Every conviction-weighted position, every regime-driven ceiling, every automatic deleverage, every gain-harvesting covered call, every re-entry tranche — all of it is the architecture expressing its design.

The Wall Street model is built on trust in individuals. Achelion is built on trust in architecture. We believe the LP who understands this distinction — who recognizes that the system cannot panic, cannot fall in love with a position, cannot rationalize holding something that should be sold, cannot be tired at 2am when a geopolitical event moves markets — will find that distinction deeply reassuring.

We are not asking LPs to trust our PM's judgment in the moment. We are asking them to evaluate the quality of the architecture we built before the moment arrived.

> *Far more money has been lost on Wall Street by investors who believed in salesmen than by investors who believed in systems. We built a system. Come and inspect it.*

"Silence is trust in the architecture."

Achelion Capital Management, LLC
Flow. Illumination. Discipline. Conviction.